JMUX Replacement Project

PROJECT STATUS MEETING

October 8, 2025

Agenda

01 // INTRODUCTIONS & SAFETY MOMENT

02 // PROJECT OVERVIEW

03 // DESIGN

04 // DEPLOYMENT PLAN

05 // SCHEDULE

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01

INTRODUCTIONS & SAFETY MOMENT

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- Before Working
  - Wear Eye Protection
  - Appropriate Clothing
  - Wear Gloves
  - Check Equipment and Surroundings
  - Keep Children and Pets away from the work area

- While Working
  - Change positions while doing repetitive work
  - Bend at the knees, not the waist
  - Never put hands underneath a lawn mower
  - Ladder Safety (firmly on the ground and 3 points of contact)
  - Take Breaks

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PROJECT OVERVIEW

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Project Overview//JMUX Replacement: Private Network

Project Goal	Modernize PSEG LI's substation communication by replacing legacy JMUX with MPLS nodes to deliver a reliable, scalable and future-ready network.
Burns & McDonnell	System Integrator Engineering Detailed Design Network Architect Construction Support
Hawkeye	Construction and Installation JMUX Removal
Nokia	Network Equipment Supplier

Why it matters: Each Supplier has their role to play in the success of this project.

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Project Overview//JMUX Replacement Scope

Project Scope

SCOPE	SITES	DF RTU SITES	RADIO SHELTER SITES	MPLS SITES	NOKIA MPLS NODES
PHASE 1	49	17	6	26	31
PHASE 2	45	12	7	26	26
PHASE 3	35	7	6	22	22
TOTAL SCOPE	129	36	19	74	79

Engineering Services

- Detailed site surveys, rack elevations, power design, supplemental documents and as-builts
- Each of the substations are fully aligned with PSEG LI standards and stakeholder buy-off
- **Phase 1: 30 of 31 NY PE** sealed construction documents approved as of 10/08/2025
  - Issue for Review (IFR)
  - Issue for Construction (IFC)
  - As-Built (Record Drawings)

Procurement & Staging

- All equipment purchased by PSEGLI and shipped to Hawkeye's staging facility.
- JMUX Replacement Project equipment will be racked, powered and validated in a controlled environment for firmware consistency, connectivity and service functionality prior to field deployment.

Construction and Installation Support

- Hawkeye to provide installation, power, and cabling for all Phase 1 nodes.
- BMcD to provide on-site coordination with engineering team, technical support and commissioning oversight.

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DESIGN

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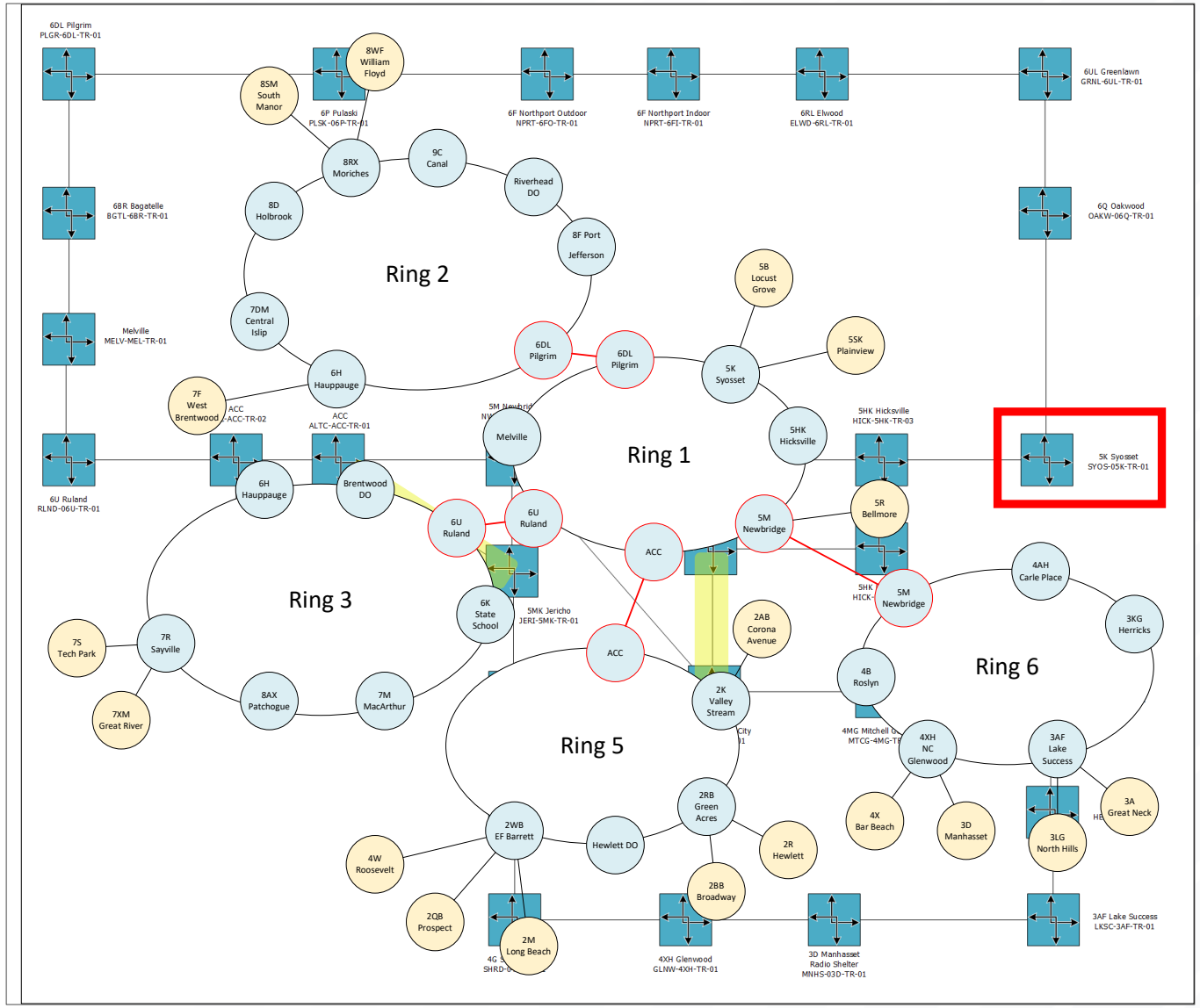
Design//Technology

Overview

- Transitioning from legacy SONET to MPLS using widely-adopted Nokia equipment
- Design emphasizes high availability
- 10 Gbps Ring Architecture
- Reduce single points of failure

Services Supported

- EMS SCADA
- DSCADA (Prod & QA)
- Teleprotection
- Two-way Radio
- Security
- DA-Radio
- MicroPAQ



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Design//MPLS Benefits

Legacy JMUX

- Aging hardware
- Single points of failure
- Fragmented monitoring
- Bandwidth constraints
- Manual Failover Process
- Limited Visibility

Nokia MPLS

- Resilient, redundant design
- Scalable (10 Gbps) bandwidth and QoS
- Real-time monitoring and control
- End-to-end visibility
- Dynamic and Automatic failover
- Enhanced Telemetry, Reporting, Logging, and Troubleshooting Capabilities

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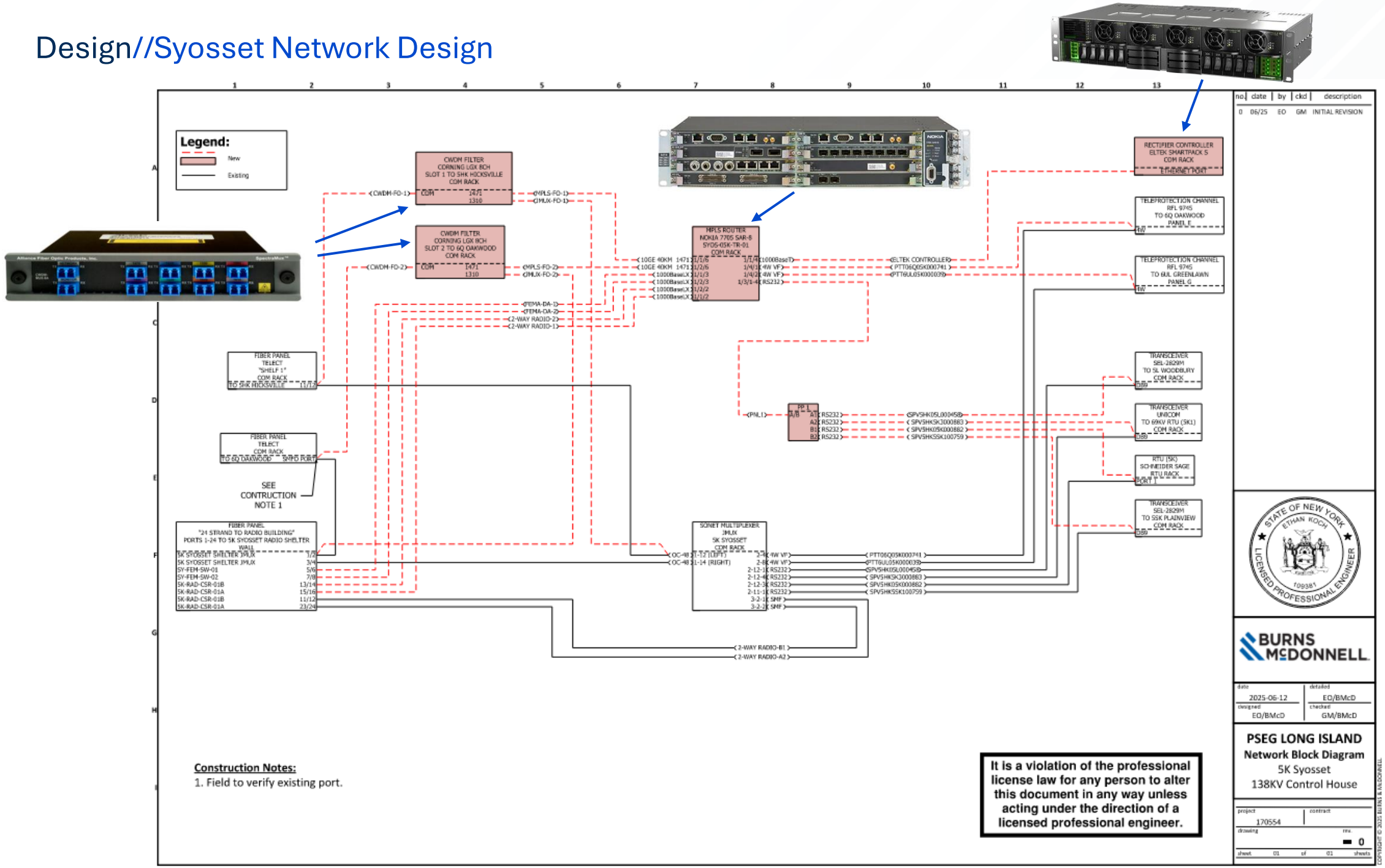
Design//Syosset Network Design



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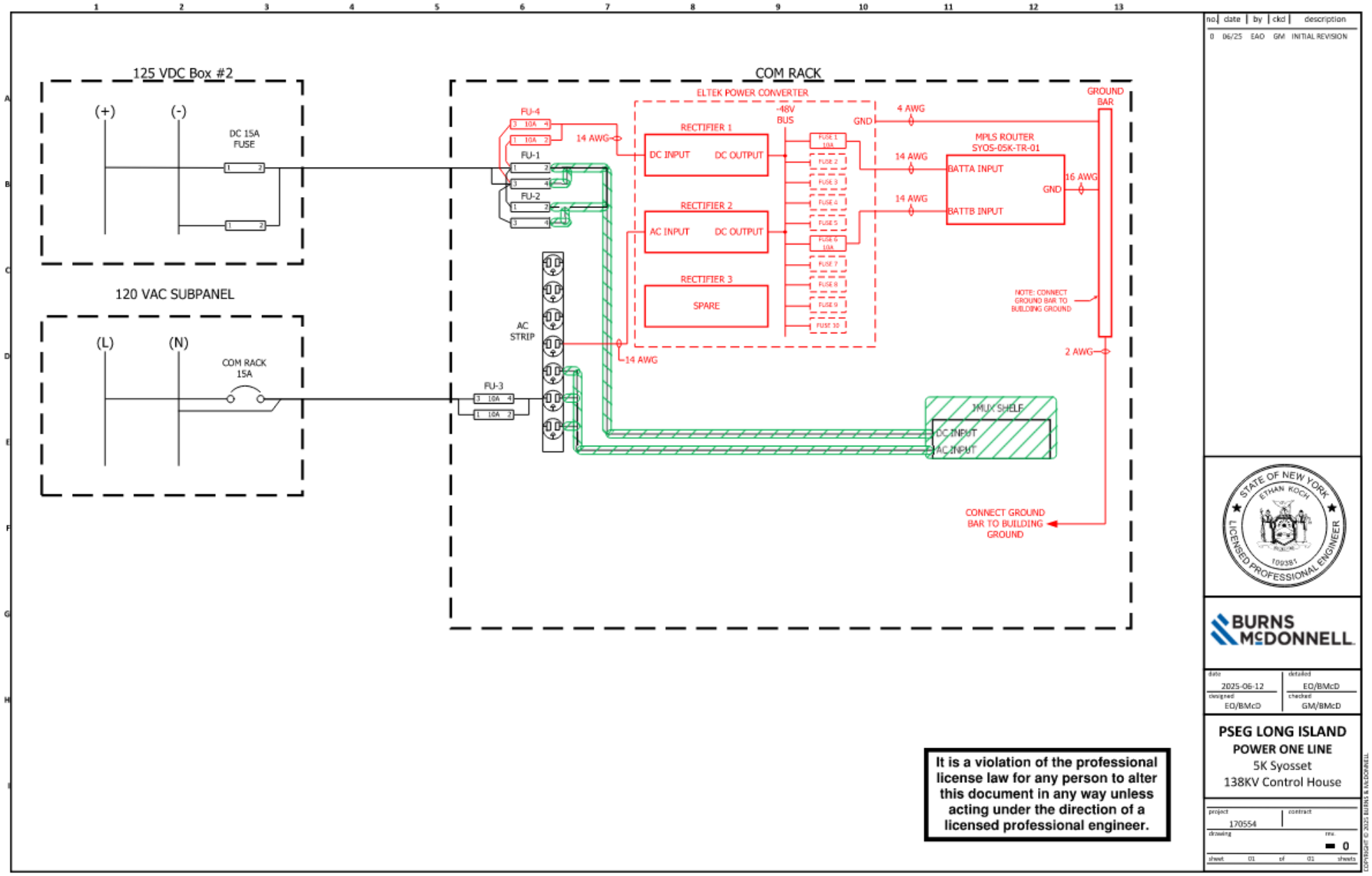
Design//Syosset Power Design



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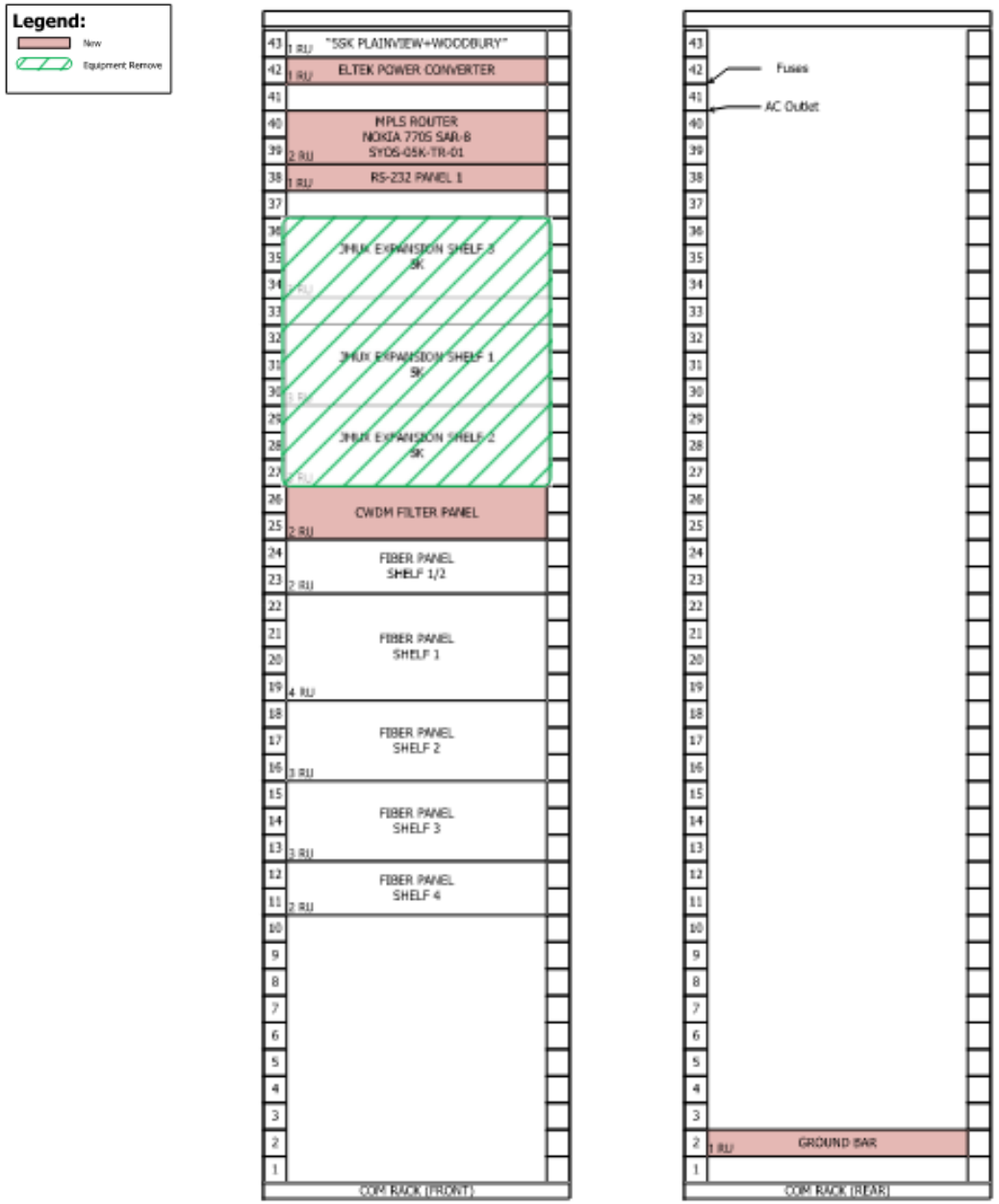
Design//Syosset Rack Elevation Example



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DEPLOYMENT PLAN

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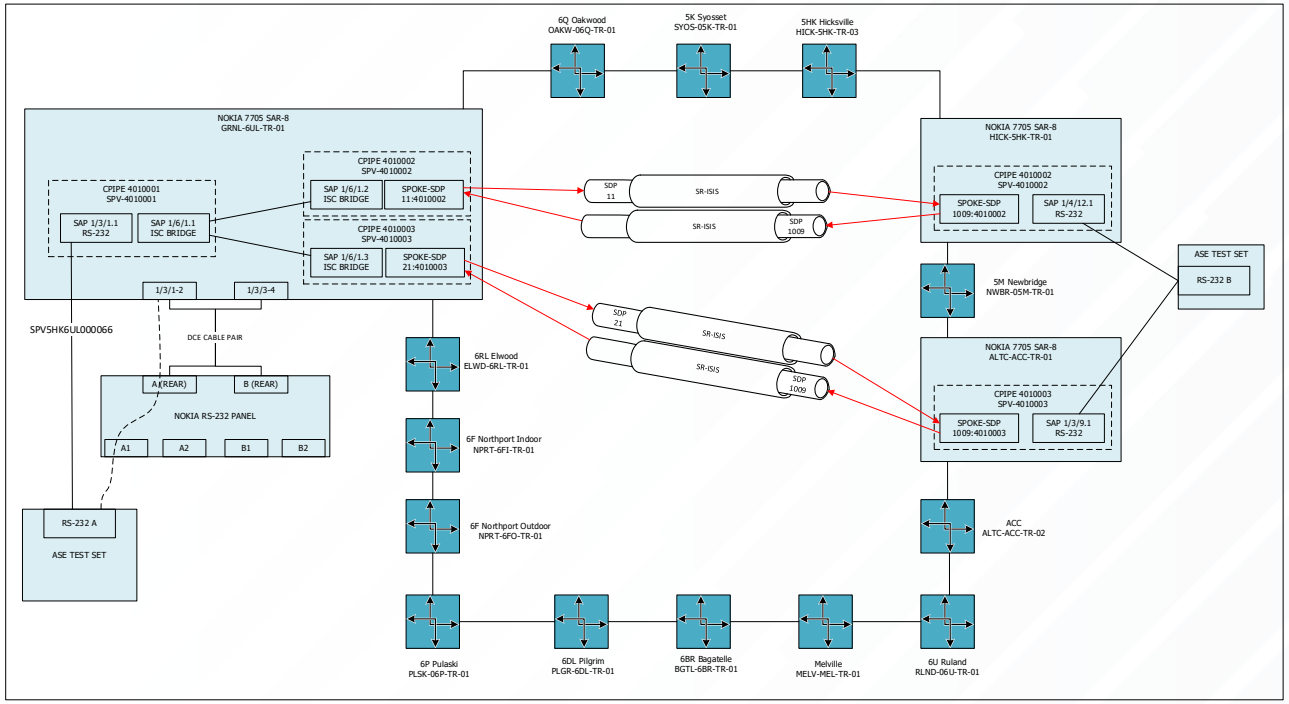
Deployment//Equipment Fully Tested and Verified

Staging	Description
POST, Assembly, and Software Updates	• Power equipment and test all hardware • Assemble equipment and make ready for field deployment • Perform all required software updates

Failure Testing	• Connect equipment as it would be in the field
	• Perform failure testing and confirm expected results are achieved
	  • Card & Module Failures
	  • Link Failures
	  • Chassis Failures
	  • Power Failures



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Deployment//Sequencing (per Station)

Mobilization	Description	Responsible Stakeholder	Present On Site	Estimated Duration/Site
Initial Installation	• Install MPLS router and CWDM • Install power feeds • Power MPLS router • Install new communication cables	Hawkeye	Hawkeye, BMcD, PSEGLI	2 days
Fiber Integration*	• JMUX OC-48 circuits & Nokia MPLS over CWDM filters • **Establish new MPLS parallel links** • Establish Nokia parallel network & connectivity	PSEGLI	PSEGLI, BMcD	0.25 day

Layer 3 Connect	• **Establish Ethernet circuits with Nokia MPLS in parallel with JMUX where possible**
	  • Security
	  • FEMA DA
	  • Two-way Radio
	  • DSCADA
	  • MicroPAQ
		BMcD, PSEGLI	BMcD	1 day

Layer 1-2 Cutover	• Cutover of relay and RTU circuits from JMUX to MPLS • Cutover any other circuits (PAR, DRD, etc.) as possible • Relays across rings may be cutover in a later phase • Only RTUs/Circuits with no dependency on other JMUX rings	PSEGLI (Relay)	PSEGLI, BMcD	2 days
JMUX Removal	• JMUX to be decommissioned and removed from site during a later phase after all circuits have been migrated • JMUX will continue to run in parallel until all Circuits/Applications have been successfully migrated	Hawkeye	Hawkeye, BMcD	1 day

Fiber Integration*: Activity will follow PSEG Change Management Process. Production JMUX will be touched in this phase to inset MPLS and will require testing scope.

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SCHEDULE

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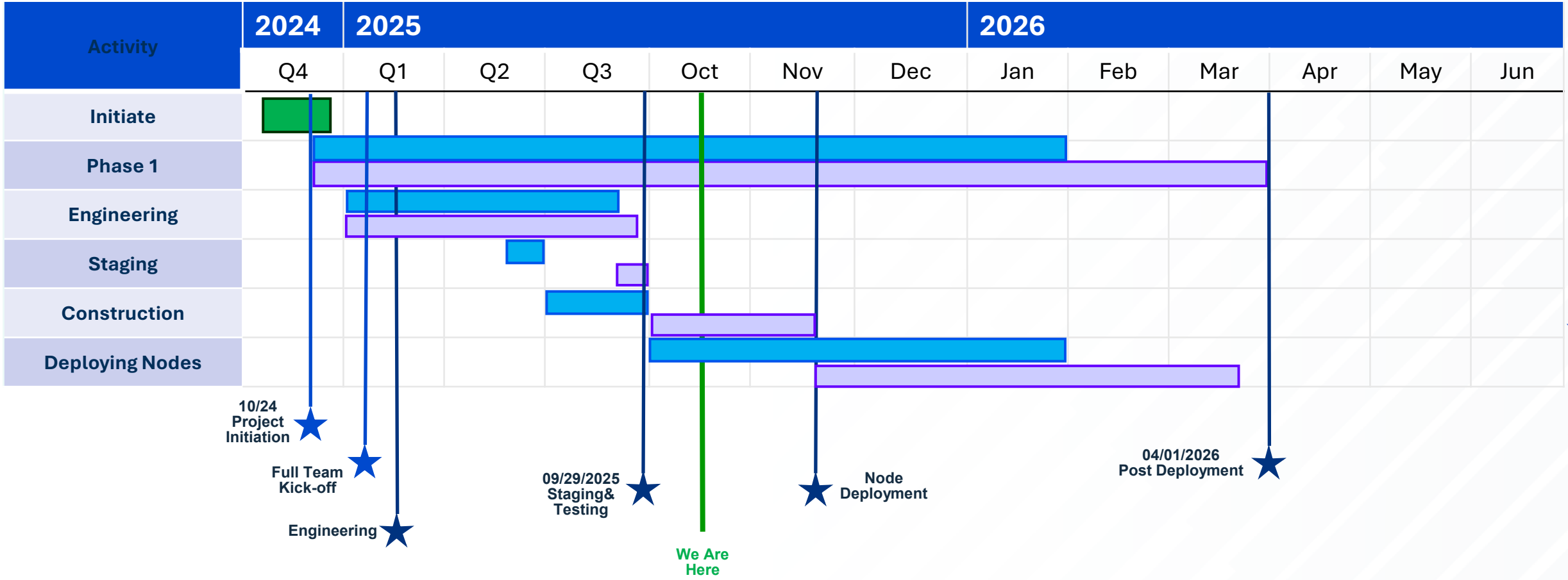
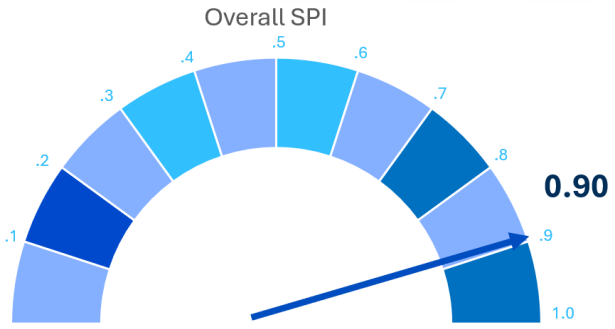
Schedule // JMUX Replacement Phase 1



Key	
Base plan	
Current	



- Baseline duration: Oct 1, 2024 → Jan 31, 2026 = **487 days**
- New duration: Oct 1, 2024 → Mar 30, 2026 = **545 days**
- Slip: **58 days** (~8.3 weeks ≈ 1.9 months)

Schedule Performance Index (SPI)

$$\text{SPI} = \frac{\textit{Baseline Duration}}{\textit{New Duration}} = \frac{487}{545} \approx \mathbf{0.90}$$

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Schedule//Phase 1 Installation Schedule

Task Name	Start	Finish
Week # 1	**Mon 10/6/25**	**Fri 10/10/25**
Melville	Mon 10/6/25	Fri 10/10/25
6BR Bagatelle	Mon 10/6/25	Fri 10/10/25
6DL Pilgrim	Mon 10/6/25	Fri 10/10/25
6P Pulaski	Mon 10/6/25	Fri 10/10/25
6P1 Pulaski Shelter	Mon 10/6/25	Fri 10/10/25
3AF Lake Success	Mon 10/6/25	Fri 10/10/25
3AF1 Lake Success Shelter	Mon 10/6/25	Fri 10/10/25
6RL Elwood	Mon 10/6/25	Fri 10/10/25
6Q Oakwood	Mon 10/6/25	Fri 10/10/25
Week # 2	**Mon 10/13/25**	**Fri 10/17/25**
5HK: Hix Hub (Primary)	Mon 10/13/25	Fri 10/17/25
5HK: Hix Hub (Secondary)	Mon 10/13/25	Fri 10/17/25
6UL Greenlawn	Mon 10/13/25	Fri 10/17/25
6UL1 Greenlawn Shelter	Mon 10/13/25	Fri 10/17/25
6U Ruland	Mon 10/13/25	Fri 10/17/25
5M Newbridge	Mon 10/13/25	Fri 10/17/25
Week # 3	**Mon 10/20/25**	**Fri 10/24/25**
Alternate Control Center (ACC)	Mon 10/20/25	Fri 10/24/25
5k Syosset	Mon 10/20/25	Fri 10/24/25
5K1 Syosset Shelter	Mon 10/20/25	Fri 10/24/25
4H East Garden City	Mon 10/20/25	Fri 10/24/25
Week # 4	**Mon 10/27/25**	**Fri 10/31/25**
4MG Mitchell Gardens	Mon 10/27/25	Fri 10/31/25
4AH Carle Place	Mon 10/27/25	Fri 10/31/25
3KG Herricks	Mon 10/27/25	Fri 10/31/25
6Fo: Northport outdoor	Mon 10/27/25	Fri 10/31/25
6Fi: Northport indoor	Mon 10/27/25	Fri 10/31/25
3D1 Manhasset Shelter	Mon 10/27/25	Fri 10/31/25
3D Manhasset	Mon 10/27/25	Fri 10/31/25
Week # 5	**Mon 11/3/25**	**Thu 11/6/25**
5MK Jericho	Mon 11/3/25	Thu 11/6/25
4XH NC Glenwood	Mon 11/3/25	Thu 11/6/25
4G Shore Road	Mon 11/3/25	Thu 11/6/25
4GH Glen Head	Mon 11/3/25	Thu 11/6/25
4B Roslyn	Mon 11/3/25	Thu 11/6/25

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Thank You